# JOEL GEORGE SAM

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## PROFILE

MSc Computer Science student with experience in machine learning and software engineering. Worked on full-stack applications, LLM-powered features, and Python ML pipelines. Skilled in API development, model evaluation, debugging, and improving reliability in production-oriented ML systems.

## EDUCATION

**MEMORIAL UNIVERSITY OF NEWFOUNDLAND** — ST. JOHN'S, NL
**Master of Science in Computer Science** (GPA : 4.0 / 4.0) — Expected Dec 2026

**SONA COLLEGE OF TECHNOLOGY** — SALEM, TN, INDIA
**Bachelor of Engineering** (Cumulative GPA: 8.81/10; First Class with Distinction) — Sep 2020 - May 2024

## WORK EXPERIENCE

**AI/ML Research Intern | C-CORE | St. John's, NL | Aug 2026 – Present**
- Develop and evaluate Python ML workflows for maritime data, working across research code and production systems.
- Investigate inference failures and improve reliability through model artifact verification, failure tracking, and reproducibility checks.
- Collaborate with researchers and engineers on Docker-based ML inference pipelines for cloud deployment.

**Machine Learning Engineer (Co-op) | DeepSense | Halifax (Remote) | Jan 2026 – May 2026**
- Built and deployed end-to-end ML pipelines to extract shoreline and vegetation boundaries from Sentinel-2 imagery.
- Retrained classification models to handle seasonal variations, improving vegetation-line detection accuracy by 14%.
- Evaluated model outputs and refined geospatial preprocessing, spatial buffering, and filtering to improve detection quality.

**IT Head | Lakecrest Independent School, St. John's | Jun 2025–Present**
- Developed and maintained features for school web applications, supporting reliable access and daily academic operations.
- Collaborated with faculty to build software solutions for school needs and mentored students in coding & problem-solving.

**Student Software Developer | Kaar Tech, Chennai | Jan 2023–Jul 2023**
- Developed a full-stack sales forecasting application using React, Node.js, and Express, integrating REST APIs and interactive dashboards.
- Built data ingestion and preprocessing pipelines, using Prophet time-series forecasting to analyze historical sales and predict future trends.

## PROJECTS

**ThineBook - AI-Powered Planner Platform | Founder & Developer | Live Demo**
- Building a full-stack planner with manual customization and NLP-based editing using text instructions.
- Integrated an LLM via the OpenAI API to convert user instructions into structured planner edits with schema validation.

**Skin Cancer Classification System**
- Deep learning-based classification model using **YOLOv5** and **TensorFlow** for dermatoscopic image analysis.
- Published **research paper** "***A Deep Learning Approach for Skin Cancer Classification using Multimodal Fusion***" at the International Conference on Green Computing for Communication Technologies (2024).

**ML-Powered Low Rate DDos Detection System**
- A Python-based preprocessing pipeline for PCAP network traffic using Scapy and Wireshark to detect LDDoS attacks.
- Trained and evaluated tuned KNN, Random Forest, and SVM models, comparing performance using F1-score.

## TECHNICAL SKILLS

**Programming& Backend:** Python, TypeScript, JavaScript, SQL, PostgreSQL, MongoDB, REST APIs
**AI&ML :** PyTorch, TensorFlow, scikit-learn, LLMs, Applied NLP, OpenAI API, ML Pipelines, ModelEvaluation
**Development & Deploymnet :** React, Next.js, Docker, AWS, Git, CI/CD, ClaudeCode, Codex